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1 import pygame, sys
2 #collaborated with my partner for the entirety of
   this project
3
4 # Define the screen size
5 SCREEN_WIDTH = 800
6 SCREEN_HEIGHT = 600
7
8 SCREEN_SIZE = (SCREEN_WIDTH, SCREEN_HEIGHT)
9
10 # Initialize Pygame
11 pygame.init()
12 pygame.font.init()
13
14
15 class Checkbox:
16     def __init__(self, surface, x, y, idnum, color=(
       230, 230, 230),
17               caption="", outline_color=(0, 0, 0
       ), check_color=(0, 0, 0),
18               font_size=22, font_color=(0, 0, 0),
19               text_offset=(28, 1), font='Ariel
       Black'):
20         self.surface = surface
21         self.x = x
22         self.y = y
23         self.color = color
24         self.caption = caption
25         self.oc = outline_color
26         self.cc = check_color
27         self.fs = font_size
28         self.fc = font_color
29         self.to = text_offset
30         self.ft = font
31
32         # identification for removal and
       reorginazation
33         self.idnum = idnum
34
35         # checkbox object
36         self.checkbox_obj = pygame.Rect(self.x, self.
       y, 12, 12)
37         self.checkbox_outline = self.checkbox_obj.
       copy()

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38
39         # variables to test the different states of
the checkbox
40         self.checked = False
41
42         # total keeps track of the total price
43         #self.total = Prices[idnum + 1] if idnum in
Prices else 0
44         self.total = 0
45
46         def _draw_button_text(self):
47             self.font = pygame.font.SysFont(self.ft, self
.fs)
48             self.font_surf = self.font.render(self.
caption, True, self.fc)
49             w, h = self.font.size(self.caption)
50             self.font_pos = (self.x + self.to[0], self.y
+ 12 / 2 - h / 2 +
51                             self.to[1])
52             self.surface.blit(self.font_surf, self.
font_pos)
53
54         def render_checkbox(self):
55             if self.checked:
56                 pygame.draw.rect(self.surface, self.color
, self.checkbox_obj)
57                 pygame.draw.rect(self.surface, self.oc,
self.checkbox_outline, 1)
58                 pygame.draw.circle(self.surface, self.cc
, (self.x + 6, self.y + 6), 4)
59
60             elif not self.checked:
61                 pygame.draw.rect(self.surface, self.color
, self.checkbox_obj)
62                 pygame.draw.rect(self.surface, self.oc,
self.checkbox_outline, 1)
63                 self._draw_button_text()
64
65         def _update(self, event_object):
66             x, y = pygame.mouse.get_pos()
67             px, py, w, h = self.checkbox_obj
68             if px < x < px + w and py < y < py + w:
69                 if self.checked:
70                     self.checked = False

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71         self.total -= Prices.get(self.idnum
    + 1, 0) # subtract prices if unchecked
72         else:
73             self.checked = True
74             self.total += Prices.get(self.idnum
    + 1, 0) # add price if its checked
75
76     def update_checkbox(self, event_object):
77         '''checked = [False, False, False, False,
    False, False, False, False]
78         total = 10 # initial price of Pizza
79
80         checked = [box.checked for box in boxes] #
    checking if each box is True or False
81
82         Prices = {1: 1.5, 2: 1, 3: 2.5, 4: 1.5, 5:
    1.5, 6: 0.5, 7: 0.25, 8: 3}
83
84         for i in range(8):
85             if checked[i] == True: #making sure to
    include initial price of the pizze as $$10
86                 topping = Prices.get(i+1)
87                 total += topping'''
88
89         if event_object.type == pygame.
    MOUSEBUTTONDOWN:
90             self.click = True
91             self._update(event_object)
92
93
94 # Create the screen
95 screen = pygame.display.set_mode((SCREEN_WIDTH,
    SCREEN_HEIGHT))
96
97 # Create two surfaces to split the frame
98 surface1 = pygame.Surface((SCREEN_WIDTH * 0.3,
    SCREEN_HEIGHT))
99 surface2 = pygame.Surface((SCREEN_WIDTH * 0.7,
    SCREEN_HEIGHT))
100
101 # Fill the surfaces with different colors
102 surface1.fill((255, 255, 255))
103 surface2.fill((0, 200, 0))
104

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105 # Blit the surfaces to the screen
106 screen.blit(surface1, (0, 0))
107 screen.blit(surface2, (SCREEN_WIDTH * 0.3, 0))
108
109 boxes = []
110
111 Prices = {1: 1.5, 2: 1, 3: 2.5, 4: 1.5, 5: 1.5, 6: 0
           .5, 7: 0.25, 8: 3}
112
113 button1 = Checkbox(screen, 50, 200, 0, caption='
Pepparoni $1.50')
114 button2 = Checkbox(screen, 50, 225, 1, caption='
Mushrooms $1.00')
115 button3 = Checkbox(screen, 50, 250, 2, caption='
Onions $2.50')
116 button4 = Checkbox(screen, 50, 275, 3, caption='
Bell Peppers $1.50')
117 button5 = Checkbox(screen, 50, 300, 4, caption='
Olives $1.50')
118 button6 = Checkbox(screen, 50, 325, 5, caption='
Basil $0.50')
119 button7 = Checkbox(screen, 50, 350, 6, caption='
Oregano $0.25')
120 button8 = Checkbox(screen, 50, 375, 7, caption='
Garlic $3.00')
121
122 boxes.append(button1)
123 boxes.append(button2)
124 boxes.append(button3)
125 boxes.append(button4)
126 boxes.append(button5)
127 boxes.append(button6)
128 boxes.append(button7)
129 boxes.append(button8)
130
131 #https://stackoverflow.com/questions/68331123/pygame
-checkbox-class-works-locally-but-not-when-imported
132 #https://pyga.me/docs/tutorials/en/intro-to-pygame.
html
133
134 #I did the above code
135 #my partner did the below code
136
137 circleX = 500

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138 circleY = 300
139 radius = 150
140 Khaki = ("Khaki1")
141
142 radius2 = 130
143 Red = ('Red')
144
145 # Update the display
146 pygame.display.flip()
147
148 baseprice = 10.00
149
150 # Run the game loop
151 while True:
152
153     # Check for events
154     for event in pygame.event.get():
155         # If the user clicks the any of the
checkboxes, highlight that checkboxe
156         if event.type == pygame.MOUSEBUTTONDOWN:
157             for box in boxes:
158                 box.update_checkbox(event)
159
160         # If the user clicks the close button, quit
the game
161         if event.type == pygame.QUIT:
162             pygame.quit()
163             sys.exit()
164
165     # total price
166     total = baseprice #used ChatGPT for logic
167     for box in boxes:
168         total+= box.total
169
170
171     # Draw the frame
172     screen.fill((0, 0, 0))
173     screen.blit(surface1, (0, 0))
174     screen.blit(surface2, (SCREEN_WIDTH * 0.3, 0))
175
176     for box in boxes:
177         box.render_checkbox()
178
179     pygame.draw.circle(screen, Khaki, (circleX,

```

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179 circleY), radius) # DRAW CIRCLE
180     # Load the background image
181     Pepperoni = pygame.image.load("Pepperoni.png")
182     Mushrooms = pygame.image.load("Mushrooms.png")
183     Onion = pygame.image.load("Onion.png")
184     Bell_Peppers = pygame.image.load("Bell Peppers.
    png")
185     Olives = pygame.image.load("Olives.png")
186     Basil = pygame.image.load("Basil.png")
187     Oregano = pygame.image.load("Oregano.png")
188     Garlic = pygame.image.load("Garlic.png")
189
190
191     #resize the images to fit
192     Pepperoni = pygame.transform.scale(Pepperoni, (
    50, 50))
193     Mushrooms = pygame.transform.scale(Mushrooms, (
    50, 50))
194     Onion = pygame.transform.scale(Onion, (50,50))
195     Bell_Peppers = pygame.transform.scale(
    Bell_Peppers, (50,50))
196     Olives = pygame.transform.scale(Olives, (50,50))
197     Basil = pygame.transform.scale(Basil, (50,50))
198     Oregano = pygame.transform.scale(Oregano, (50,50
    ))
199     Garlic = pygame.transform.scale(Garlic, (50,50))
200
201
202     # Blit the background image to the screen
203     #screen.blit(background_image, (0, 0))
204     pygame.draw.circle(screen, Red, (circleX,
    circleY), radius2)
205
206     font = pygame.font.SysFont('Arial', 25)
207     font.set_bold(True)
208     pygame.draw.rect(screen, "red", (50, 425, 165,
    50))
209     # have to display total now
210     text_surface = font.render('Total : $' + str(
    total), True, 'Black') # Generated by ChatGPT for
    syntax
211     screen.blit(text_surface, (70, 435))
212
213     if button1.checked:

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```
214         screen.blit(Pepperoni, (circleX - 50,
circleY - 50))
215
216         if button2.checked:
217             screen.blit(Mushrooms, (circleX + 0, circleY
- 50))
218
219         if button3.checked:
220             screen.blit(Onion, (circleX - 100, circleY
- 50))
221
222         if button4.checked:
223             screen.blit(Bell_Peppers, (circleX - 50,
circleY - 0))
224
225         if button5.checked:
226             screen.blit(Olives, (circleX - 50, circleY
- 100))
227
228         if button6.checked:
229             screen.blit(Basil, (circleX - 0, circleY - 0
))
230
231         if button7.checked:
232             screen.blit(Oregano, (circleX - 100, circleY
- 0))
233
234         if button8.checked:
235             screen.blit(Garlic, (circleX - 0, circleY -
100))
236
237         # Update the display
238         pygame.display.flip()
239
240
```